



Pierre Gassendi's "atomic" theory was ahead of its time.



Otto von Guericke's evacuated copper hemispheres were sealed together by nothing more than a smear of grease, yet two teams of eight horses were unable to pull them apart.

pumping, rather than suction. Von Guericke published descriptions of his experiments in 1672 in *Experimenta Nova* (*New Experiments*), which also contained illustrations of his vacuum pump design.

While physical investigators debated the nature of matter, **biologists questioned the origins of life**. Many took the view that life could arise spontaneously. In 1651, English physician **William Harvey**, who had previously described the circulation of blood (see 1628–30), maintained that **animals could only originate from eggs**. After studying chickens, he set out to find the mammalian "egg." As royal physician, Harvey was granted access to the king's fallow deer for his studies. He examined pregnant animals killed ever closer to the point of copulation in the hope of tracing the source of the egg. Harvey did not know that embryonic development in deer is naturally delayed for up to eight weeks after fertilization—so he wrongly concluded that the egg arose spontaneously in the womb. Mammalian eggs were not found until the 1800s, when ovaries were examined with microscopes.

ENGLISH PHYSICIAN NICHOLAS CULPEPER (1616–54) combined interests in medicine and botany to publish *The English Physician* in 1652—a book that integrated herbal medicine with astrology—followed by *Complete Herbal* in 1653, which catalogued the **medicinal uses of plants**. Many of the plants he listed are still in use in medicine today—for example, foxglove (*Digitalis*) for heart conditions. Culpeper's work included descriptions of many remedies that had been kept a secret up to that time.

In 1653, Blaise Pascal published the results of his exhaustive studies in the physics of liquids in *Treatise on the Equilibrium of*



Probability

As the Pascal–Fermat theory tells a gambler, the probability of throwing a double 6 with two dice is 1 in 36.

Liquids. This publication contained the idea that later came to be known as **Pascal's Law**: an incompressible liquid's pressure in a small closed system is equal in all directions.

“THE HERBS OUGHT TO BE DISTILLED WHEN THEY ARE IN THEIR GREATEST VIGOR... SO OUGHT THE FLOWERS.”

Nicholas Culpeper, English botanist, from *The English Physician*, 1652



In 1654, four years after creating a vacuum pump to extract air from two sealed hemispheres, **Otto von Guericke** staged one of the most **dramatic public experiments of all time** in front of an audience at his hometown Magdeburg, Germany. Guericke sealed two copper hemispheres with grease, evacuated them using the pump, and suspended them between two teams of eight horses. Such was the strength of the air pressing in on the two copper hemispheres, that the horses could not pull them apart despite their best efforts. This astounded the assembled audience, and helped Guericke prove the **power of the vacuum**.

Meanwhile, the gambling habits of French nobleman Antoine Gombaud (1607–84) were about to help open a new field of mathematics. Gombaud questioned the profitability of a certain strategy in a game of dice, so he called upon the assistance of mathematician **Blaise Pascal** to explore the subject. Pascal initiated correspondence with his contemporary **Pierre de Fermat** (see 1635–37) to solve the

Complete Herbal

This illustration of medicinal plants is from an 1850 edition of English physician Nicholas Culpeper's *Complete Herbal*.

problem. This collaboration resulted in the **formalization of the principles of probability**. After hearing about the Pascal–Fermat exchange, Dutch scholar Christiaan Huygens published the first book on probability in 1657. Because of the widespread enthusiasm for gambling, probability theory became popular among those who took the trouble to understand it.



BLAISE PASCAL
(1623–62)

Born in Clermont-Ferrand in France, Blaise Pascal was a child prodigy who was tutored by his tax-collector father. While still in his teens, he solved a complex mathematical problem and invented a mechanical calculating machine. He helped form the basis of probability theory and the physics of hydraulics.

1651 William Harvey asserts that all animals develop from eggs

December 1652 Danish physician Thomas Bartholin names and describes the lymphatic system

1653 Culpeper publishes *Complete Herbal*

1653 Francis Glisson describes the anatomy of the liver

May 8, 1654 Otto von Guericke demonstrates the power of the vacuum in the Magdeburg demonstration

1652 Nicholas Culpeper publishes *The English Physician*

1653 Pascal publishes studies on hydrodynamics and hydraulics

1653 Pascal publishes a treatise on an arithmetical triangle, later known as *Pascal's Triangle*

July 1654 Frenchmen Pierre de Fermat and Blaise Pascal develop probability theory